

Title Page

- Lab Title: Advanced Linux Commands, Standard Streams and Controlled Evidence Transfer
- Student Name: Hauwa Haruna Yahaya.
- Student ID: FWSD/2025/11219.
- Course Name: Basic Computer Skills for Digital Forensics
- Instructor Name: Aminu Idris.
- Date of Submission: 05/9/2026

Executive Summary

This forensic laboratory exercise focused on data carving techniques essential for recovering digital evidence when file system metadata is unavailable or corrupted.

Objectives

The primary objective of this laboratory exercise was to develop practical proficiency in data carving techniques for digital forensic investigations where file system metadata is missing, damaged, or unreliable.

Tools And Resources Used

The Sleuth Kit

Kali linux

binwalk

Methodology

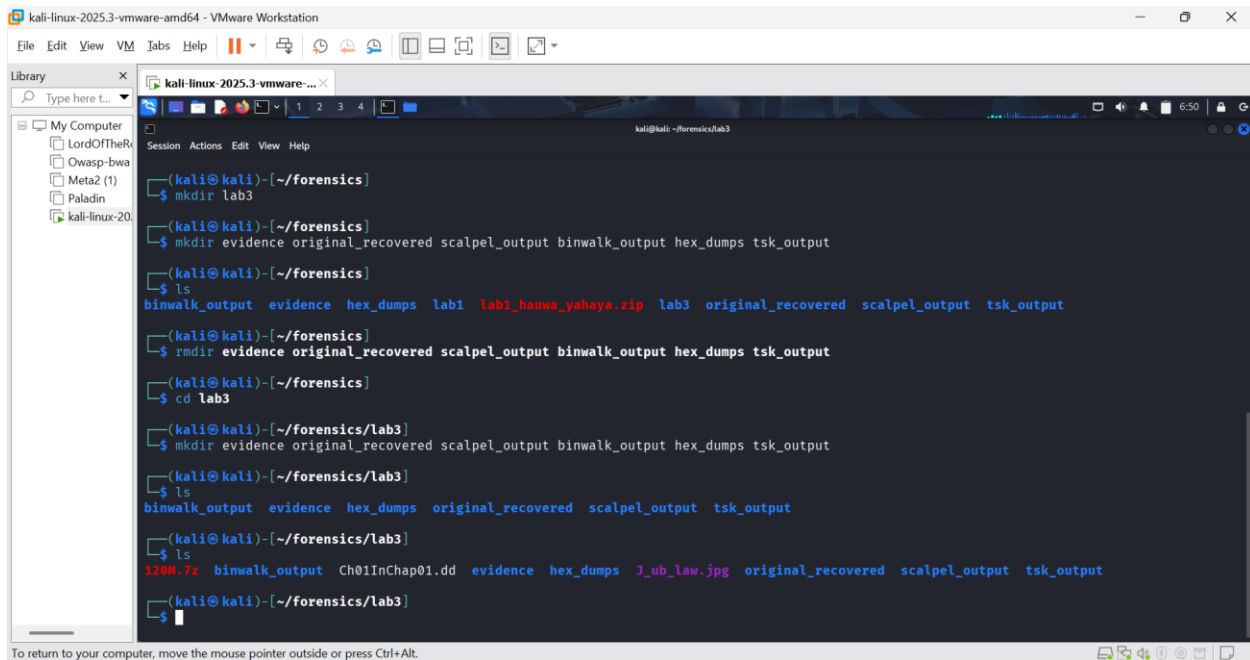


FIGURE 1: I created a dedicated folder for the lab

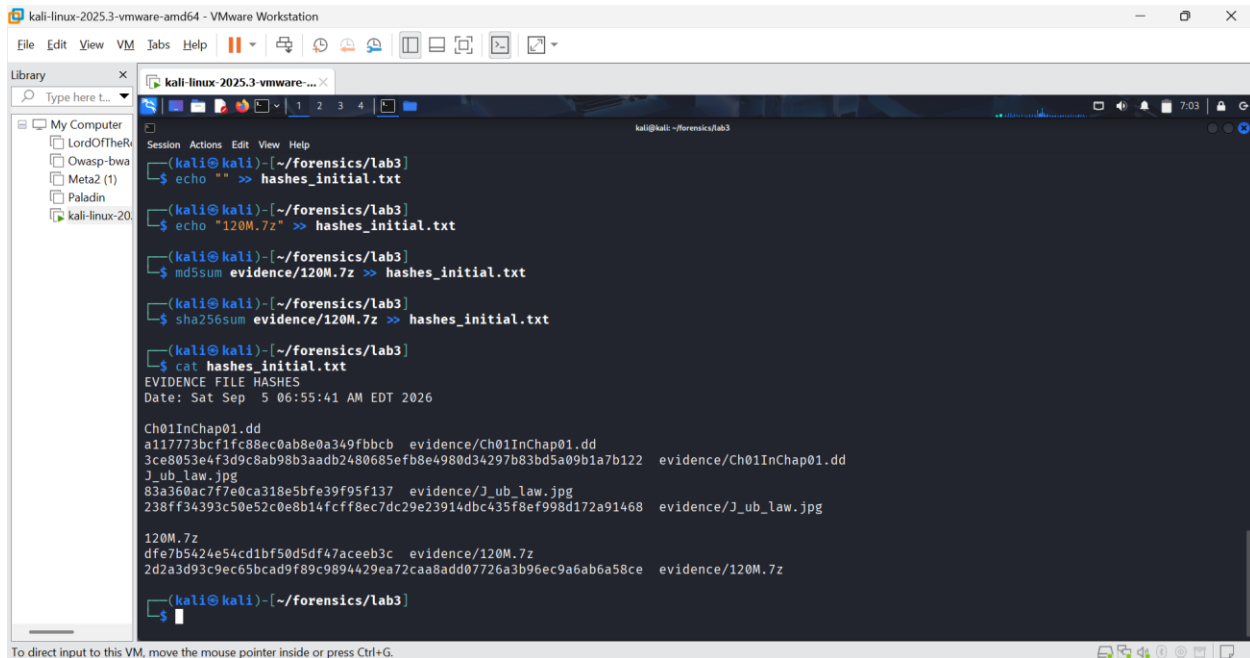
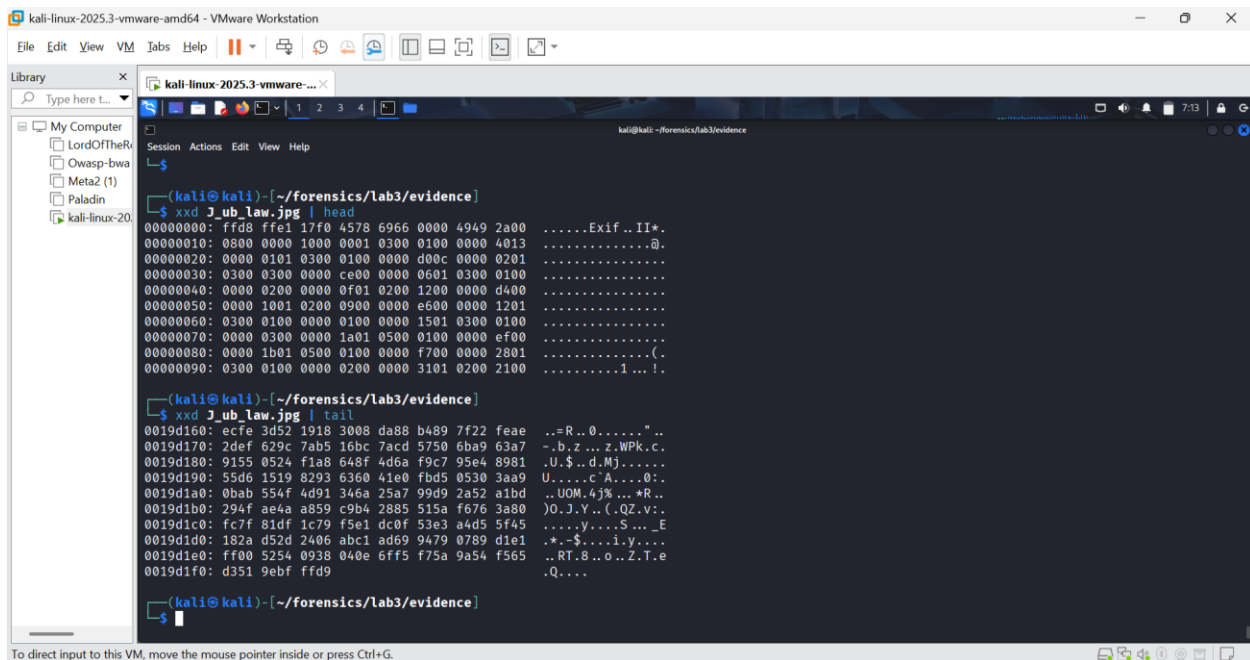


FIGURE 2: I calculated the initial hashes.

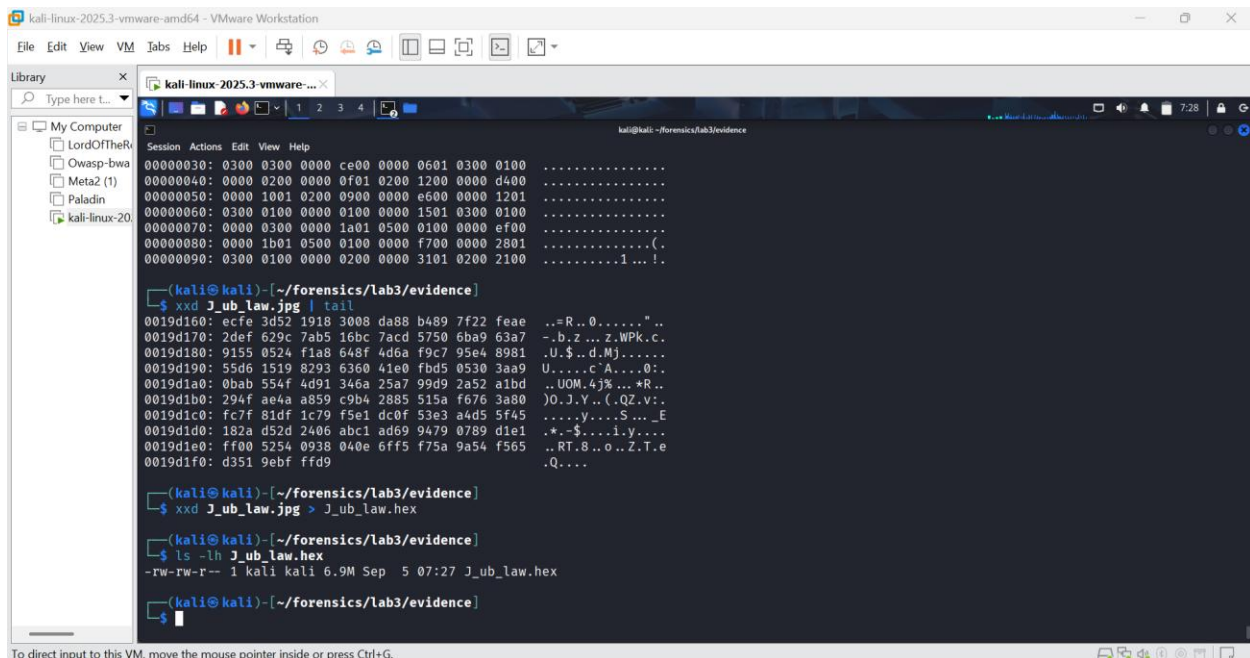
Part A - File Signatures and Hex Analysis



```
kali@kali:~/forensics/lab3/evidence$ xxd J_ub_law.jpg | head
00000000: ffd8 ffe1 17f0 4578 6966 0000 4949 2a00 .....Exif..II*.
00000010: 0800 0000 1000 0001 0100 0100 0000 4013 .....0.
00000020: 0000 0101 0300 0100 0000 d00c 0000 0201 .....
00000030: 0300 0300 0000 ce00 0000 0601 0300 0100 .....
00000040: 0000 0200 0000 0f01 0200 1200 0000 d400 .....
00000050: 0000 1001 0200 0900 0000 e600 0000 1201 .....
00000060: 0300 0100 0000 0100 0000 1501 0300 0100 .....
00000070: 0000 0300 0000 1a01 0500 0100 0000 ef00 .....
00000080: 0000 1b01 0500 0100 0000 f700 0000 2801 .....
00000090: 0300 0100 0000 0200 0000 3101 0200 2100 .....1 ...!.

kali@kali:~/forensics/lab3/evidence$ xxd J_ub_law.jpg | tail
0019d160: ecfe 3d52 1918 3008 da88 b489 7f22 feae ..=R..0....." ..
0019d170: 2def 629c 7ab5 16bc 7acd 5750 6ba9 63a7 ..b.z...z.WPk.C.
0019d180: 9155 0524 f1a8 648f 4d6a f9c7 95e4 8981 ..U.$..d.Mj.....
0019d190: 55d6 1519 8293 6360 41e0 fbd5 0530 3aa9 U.....c'A....0:
0019d1a0: 0bab 554f 4d91 346a 25a7 99d9 2a52 a1bd ..UOM.4j%...*R..
0019d1b0: 294f ae4a a859 c9b4 2885 515a f676 3a80 )O.J.Y..(.QZ.V:
0019d1c0: fc7f 81df 1c79 f5e1 dc0f 53e3 a4d5 5f45 ....Y....S..._E
0019d1d0: 182a d52d 2406 abc1 ad69 9479 0789 d1e1 ..*-$....i.y....
0019d1e0: ff00 5254 0938 040e 6ff5 f75a 9a54 f565 ..RT.B...o..Z.T.e
0019d1f0: d351 9ebf ffd9 ..Q....
```

FIGURE 3: I used xxd command to identify the jpeg header and tail. I identified ffd8 ffe1 as the header and ffd9 as the tail.



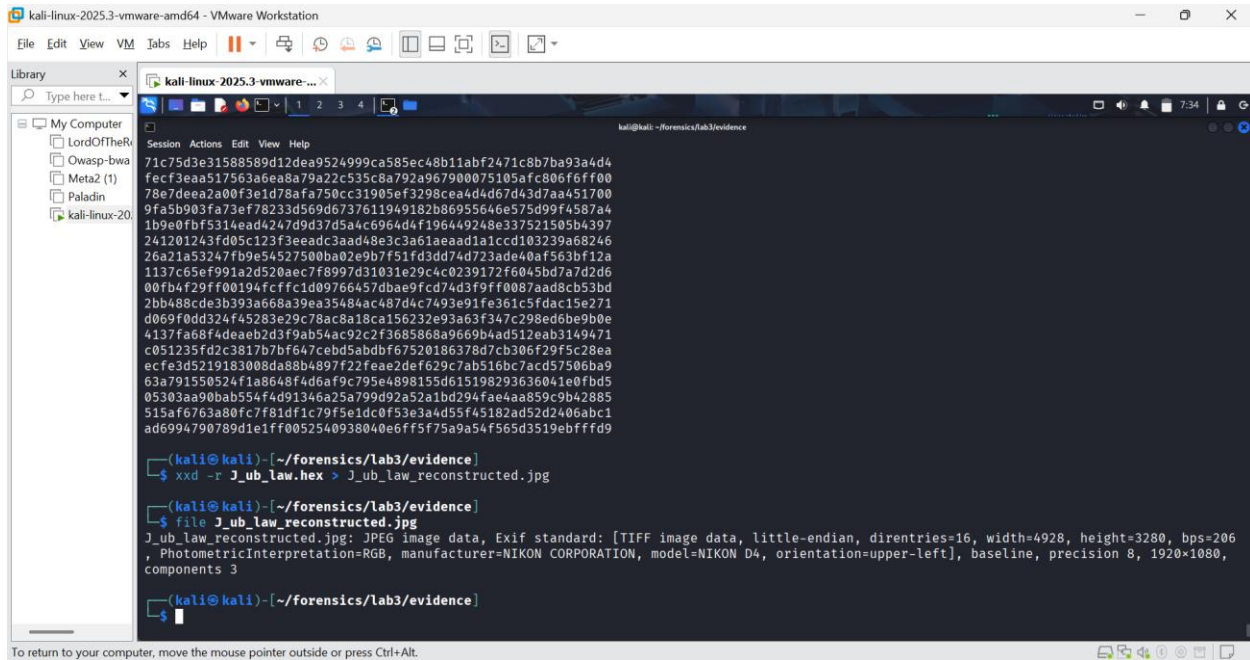
```
kali@kali:~/forensics/lab3/evidence$ xxd J_ub_law.jpg | tail
0019d160: ecfe 3d52 1918 3008 da88 b489 7f22 feae ..=R..0....." ..
0019d170: 2def 629c 7ab5 16bc 7acd 5750 6ba9 63a7 ..b.z...z.WPk.C.
0019d180: 9155 0524 f1a8 648f 4d6a f9c7 95e4 8981 ..U.$..d.Mj.....
0019d190: 55d6 1519 8293 6360 41e0 fbd5 0530 3aa9 U.....c'A....0:
0019d1a0: 0bab 554f 4d91 346a 25a7 99d9 2a52 a1bd ..UOM.4j%...*R..
0019d1b0: 294f ae4a a859 c9b4 2885 515a f676 3a80 )O.J.Y..(.QZ.V:
0019d1c0: fc7f 81df 1c79 f5e1 dc0f 53e3 a4d5 5f45 ....Y....S..._E
0019d1d0: 182a d52d 2406 abc1 ad69 9479 0789 d1e1 ..*-$....i.y....
0019d1e0: ff00 5254 0938 040e 6ff5 f75a 9a54 f565 ..RT.B...o..Z.T.e
0019d1f0: d351 9ebf ffd9 ..Q....

kali@kali:~/forensics/lab3/evidence$ xxd J_ub_law.jpg > J_ub_law.hex

kali@kali:~/forensics/lab3/evidence$ ls -lh J_ub_law.hex
-rw-rw-r-- 1 kali kali 6.9M Sep  5 07:27 J_ub_law.hex

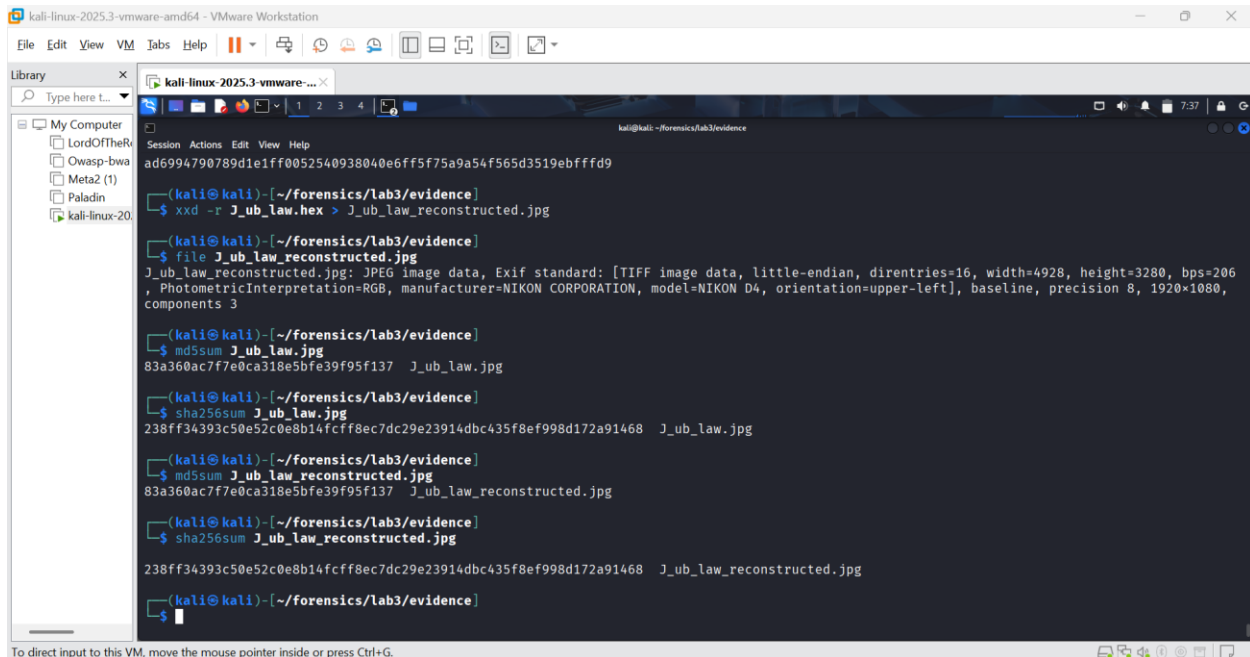
kali@kali:~/forensics/lab3/evidence$
```

FIGURE 4: I created a full hex dump of the image



To return to your computer, move the mouse pointer outside or press Ctrl+Alt.

FIGURE 5: I reconstructed the image using reverse hex dump functionality



To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

FIGURE 6: I verified the original image against the reconstructed image

Part B - Examine Embedded Metadata-Like Content

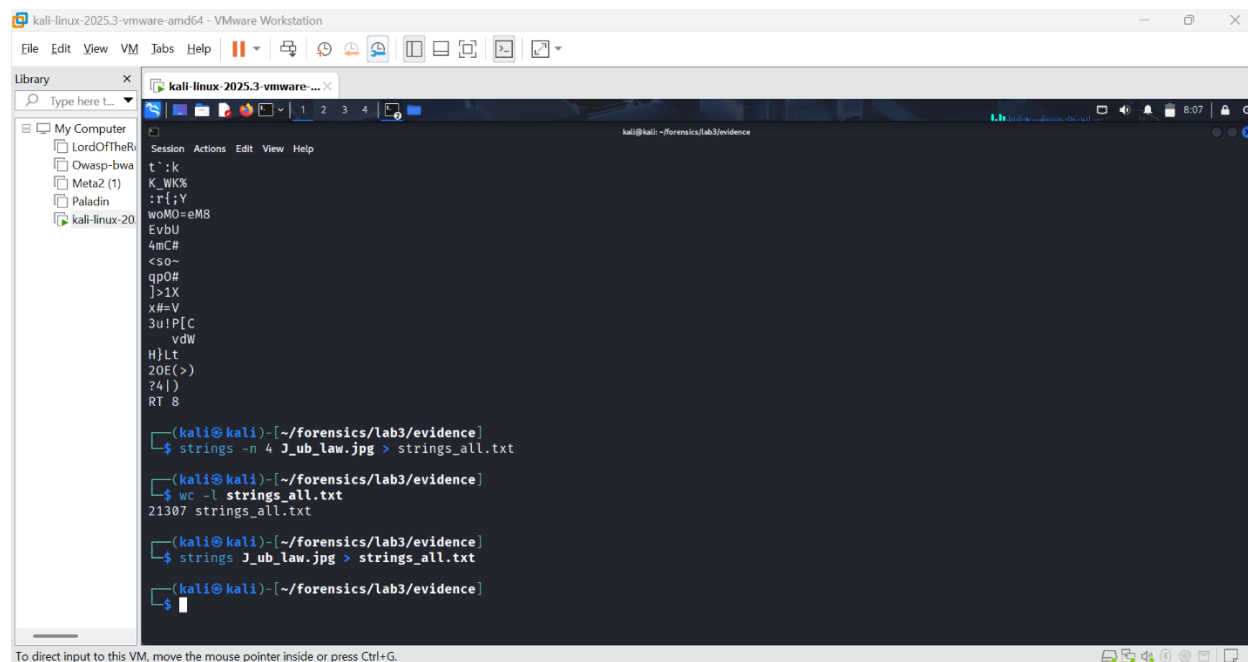


FIGURE 7: I extracted all readable strings

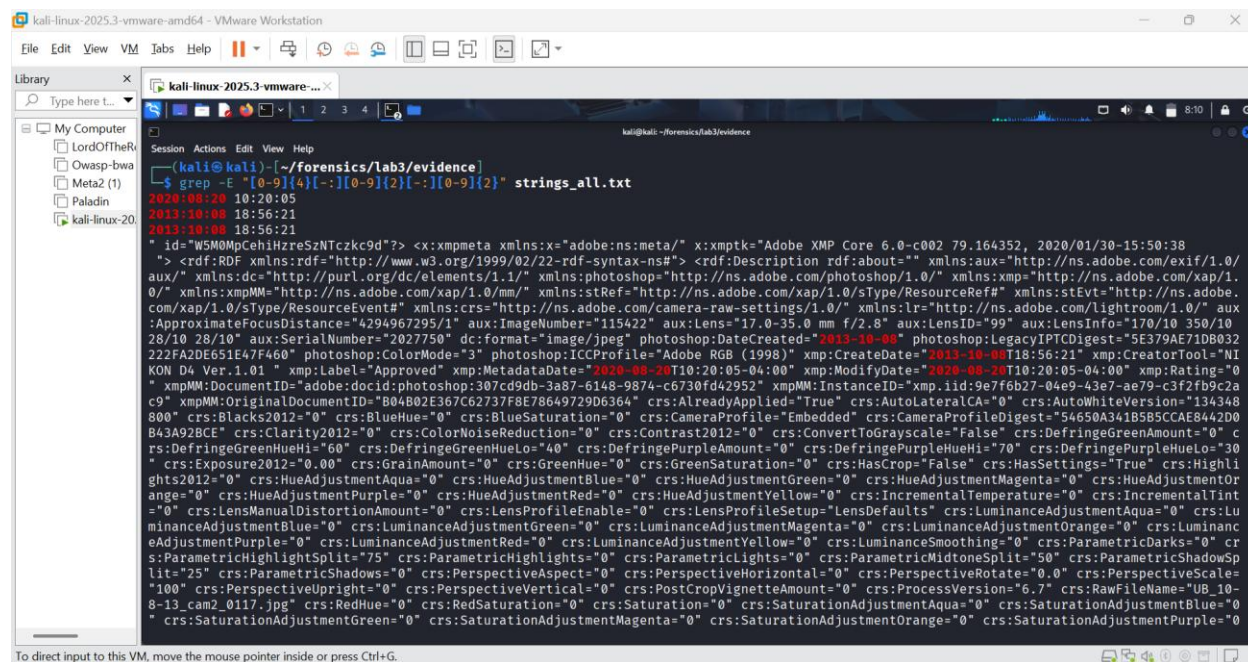


FIGURE 8: I searched for dates

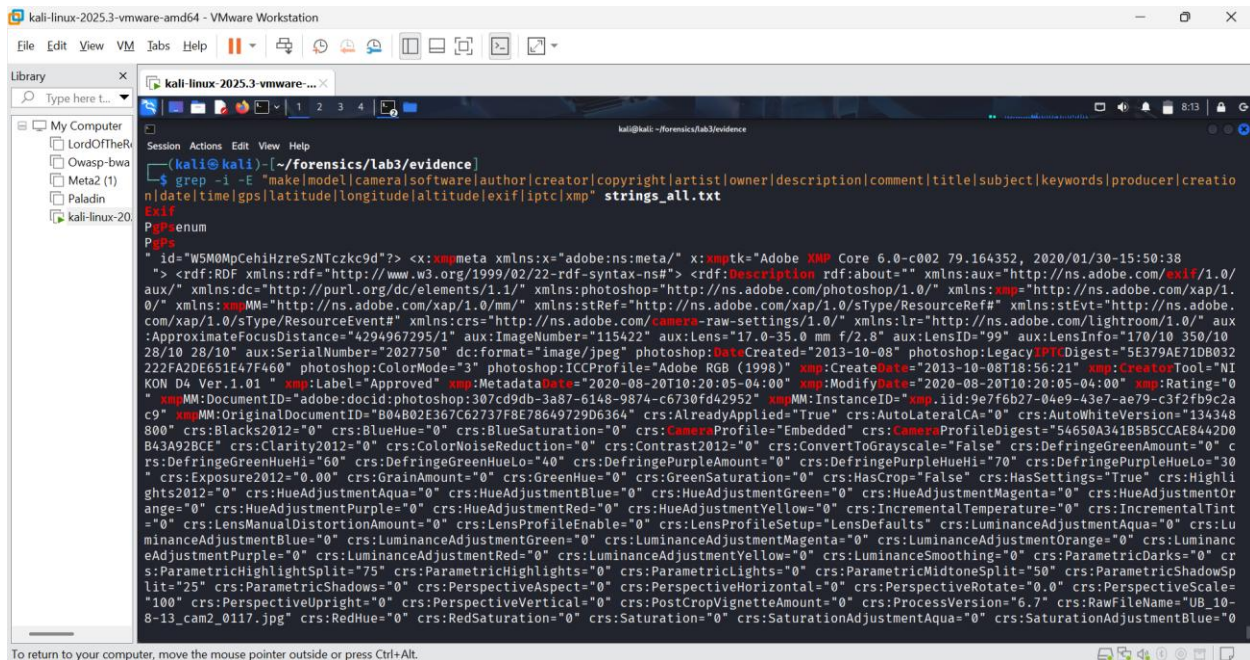


FIGURE 9: I searched for metadata related strings.

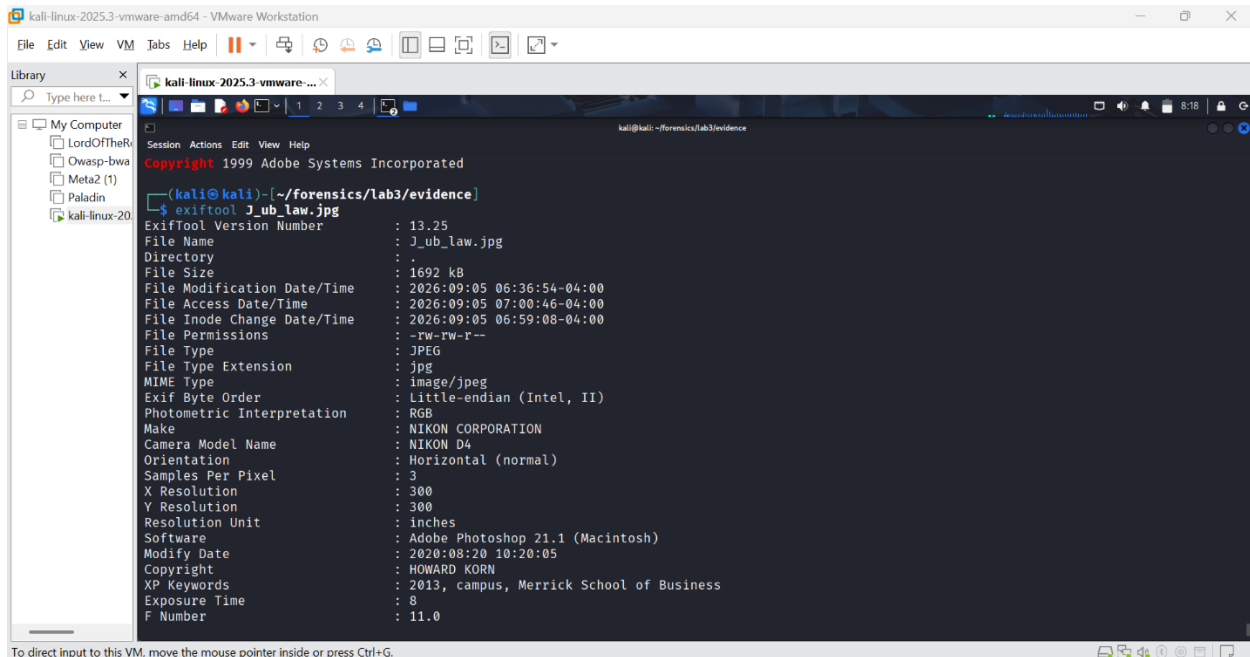


FIGURE 10: I used exiftool to extract standard metadata

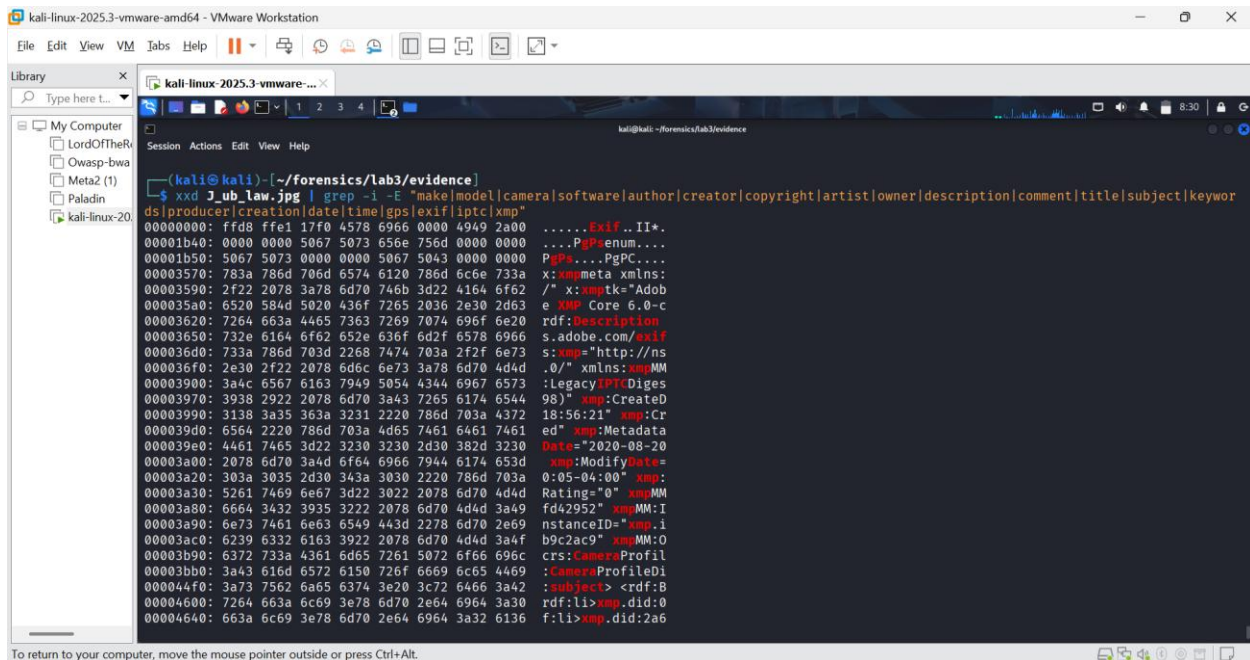


FIGURE 11: I used xxd to search for common metadata

Part C - Direct Data-Unit Examination with The Sleuth Kit

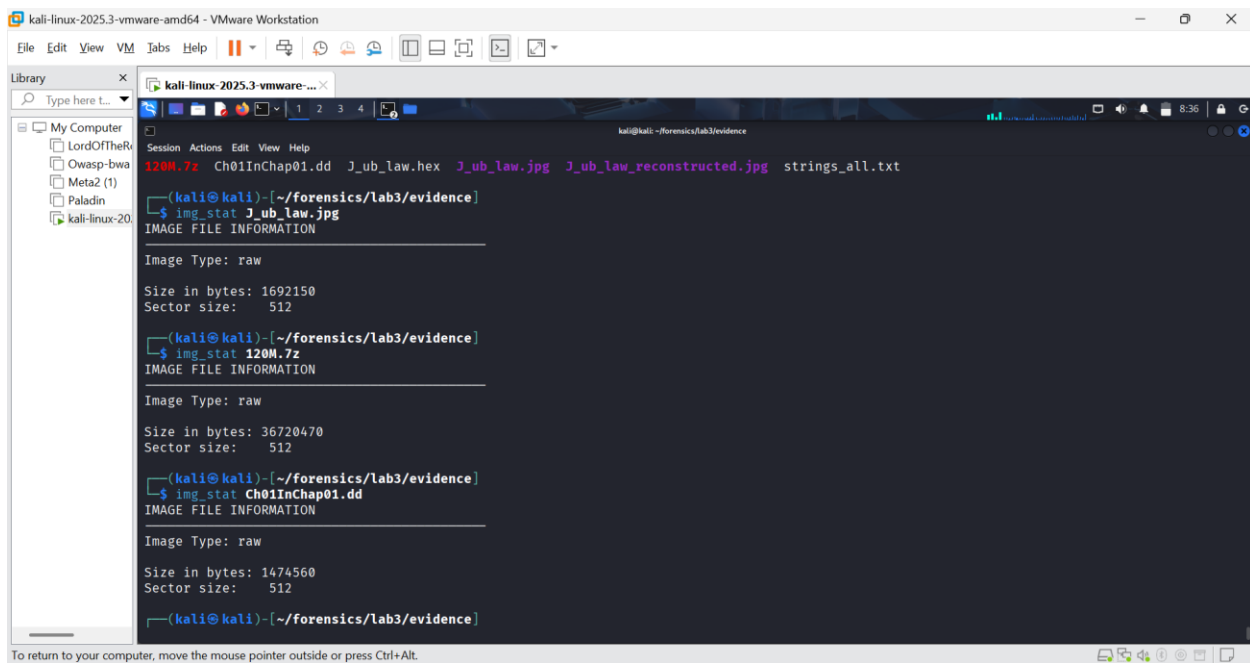


FIGURE 12: Image Information

```

(kali㉿kali)-[~/forensics/lab3/evidence]
$ mmls J_ub_law.jpg

(kali㉿kali)-[~/forensics/lab3/evidence]
$ mmls 120M.7z

(kali㉿kali)-[~/forensics/lab3/evidence]
$ mmls Ch01InChap01.dd

```

FIGURE 13: The `mmls` command returned empty for all three files because `J_ub_law.jpg` is a JPEG image file, not a disk image, `120M.7z` is a compressed archive (needs extraction), `Ch01InChap01.dd` is a FAT12 filesystem image without a partition table.

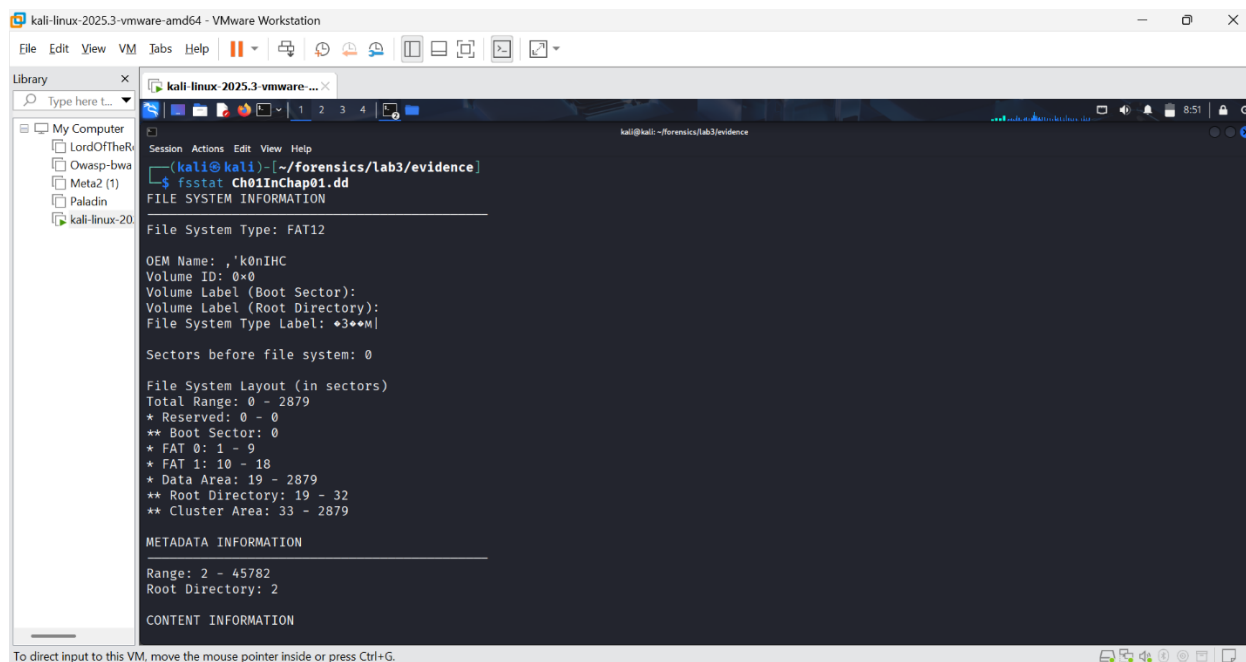


FIGURE 14: File system information for `Ch01InChap01.dd`

```

(kali㉿kali)-[~/forensics/lab3/evidence]
$ fls Ch01InChap01.dd
r/r 5: Client Info.mdb
r/r * 8: Billing Letter.doc
r/r * 11: confirmation.txt
r/r 13: Income.xls
r/r * 15: letter1.txt
r/r * 17: Regrets.doc
v/v 45779: $MBR
v/v 45780: $FAT1
v/v 45781: $FAT2
V/V 45782: $OrphanFiles

```

FIGURE 15: File list for `Ch01InChap01.dd`

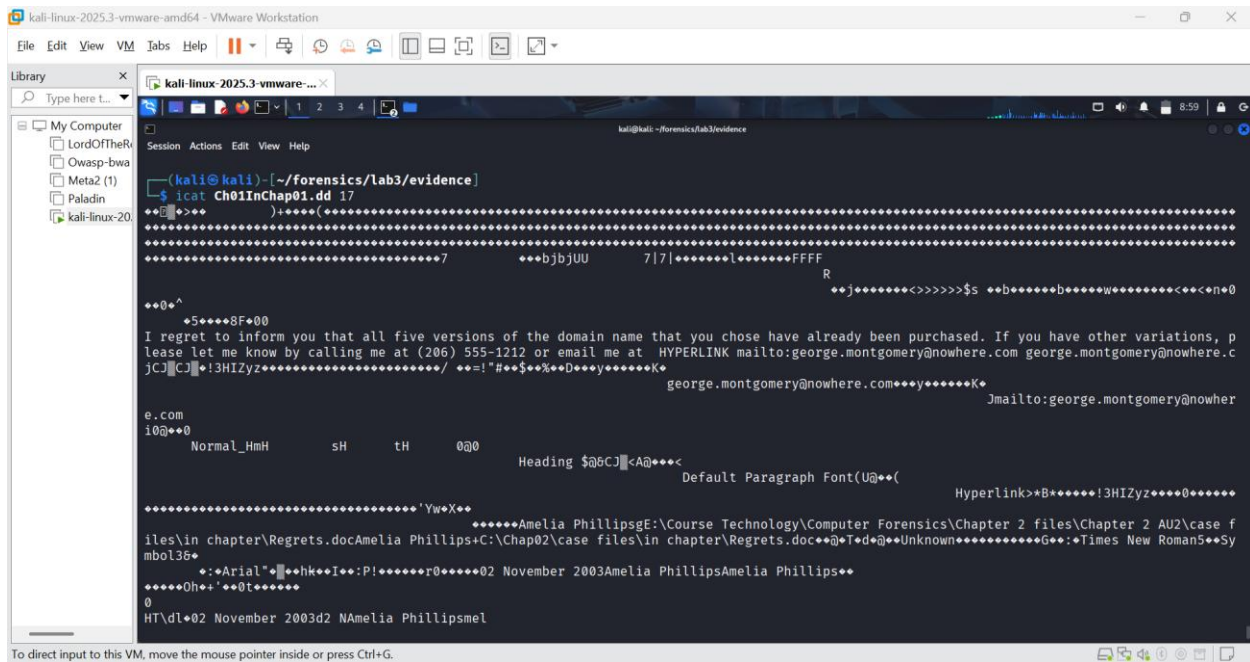


FIGURE 16: I recovered a deleted file with inode number 17.

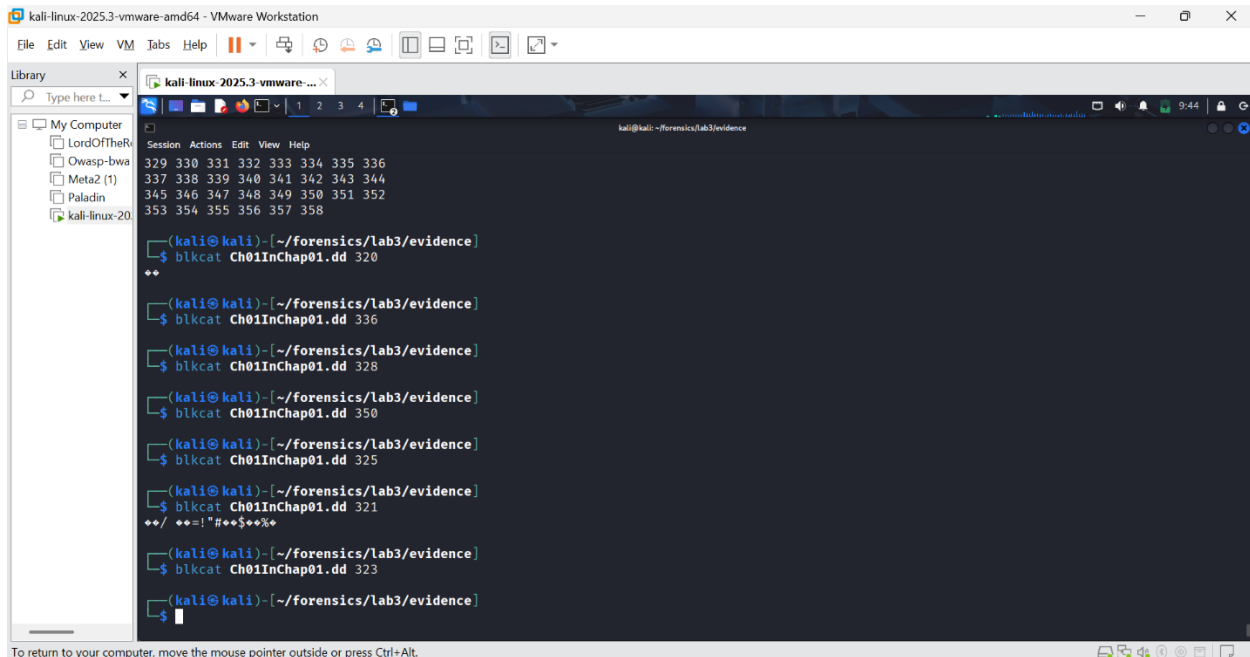


FIGURE 17: I extracted the contents of a given data unit

Part D - Embedded File Discovery with Binwalk

The instructor-provided File_carving.docx was not available in the evidence package. However, binwalk analysis was completed on available evidence files (J_ub_law.jpg and 120M.7z).

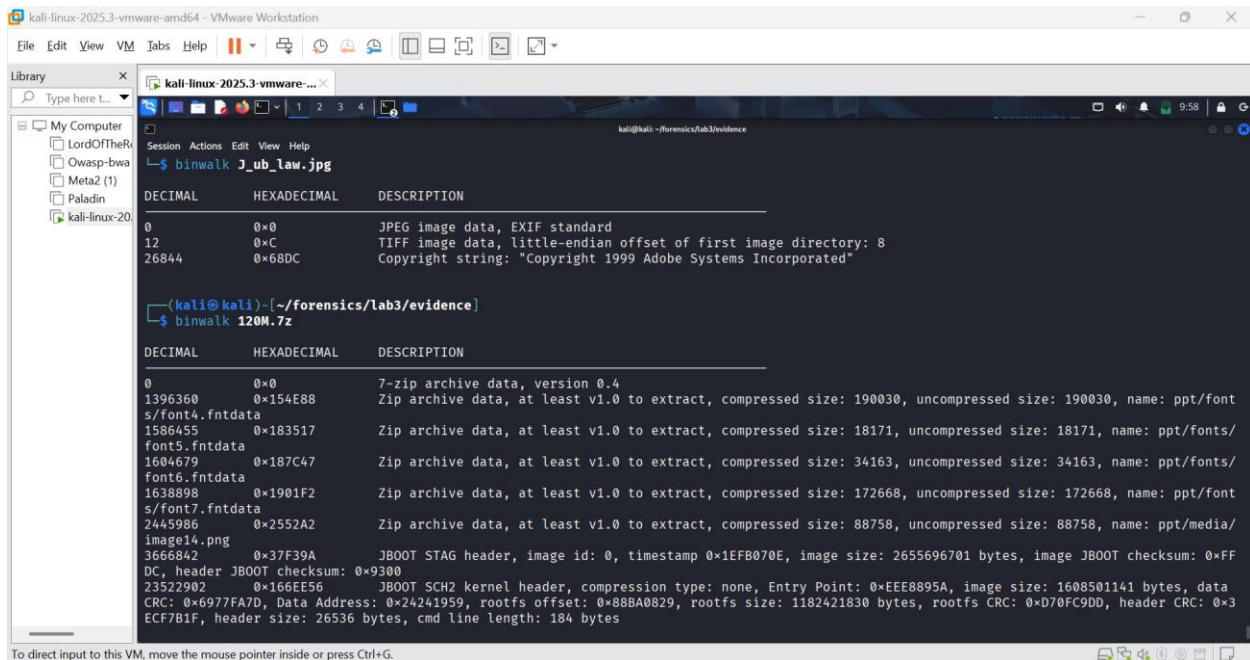


FIGURE 18: binwalk scan showing embedded signatures in JPEG file and 120M.7z

Part E - Prepare the USB Image

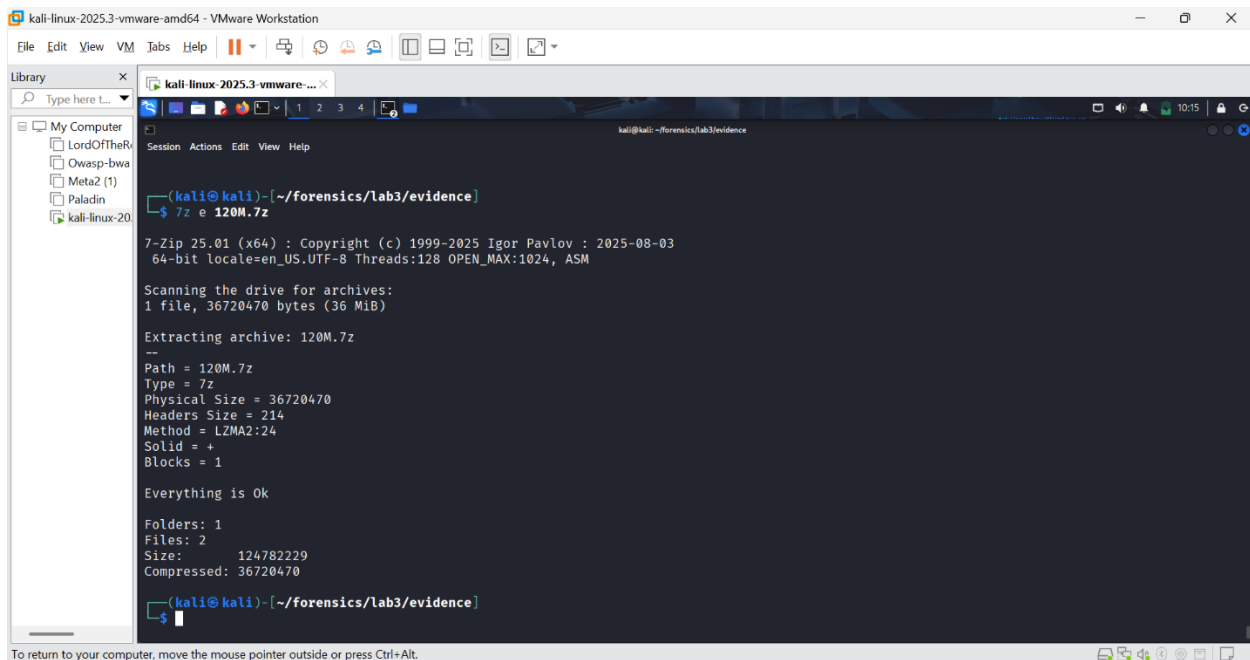


FIGURE 19: I extracted 120M.7z

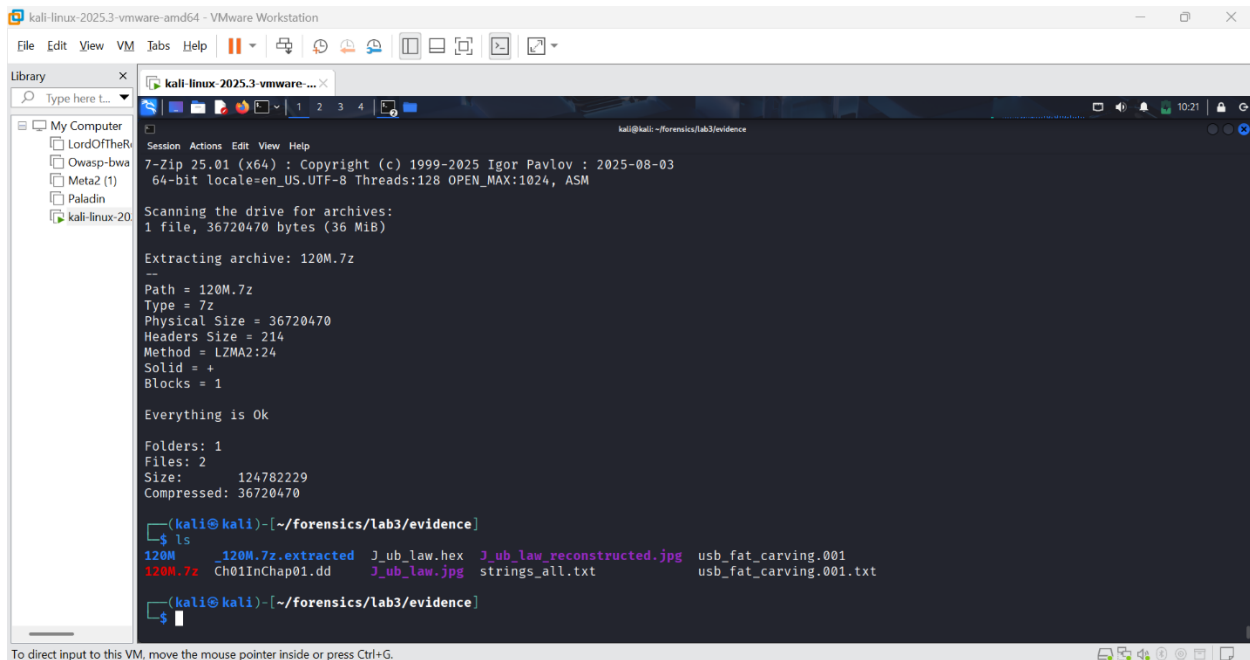


FIGURE 20: I identified `usb_fat_carving.001` as the USB image

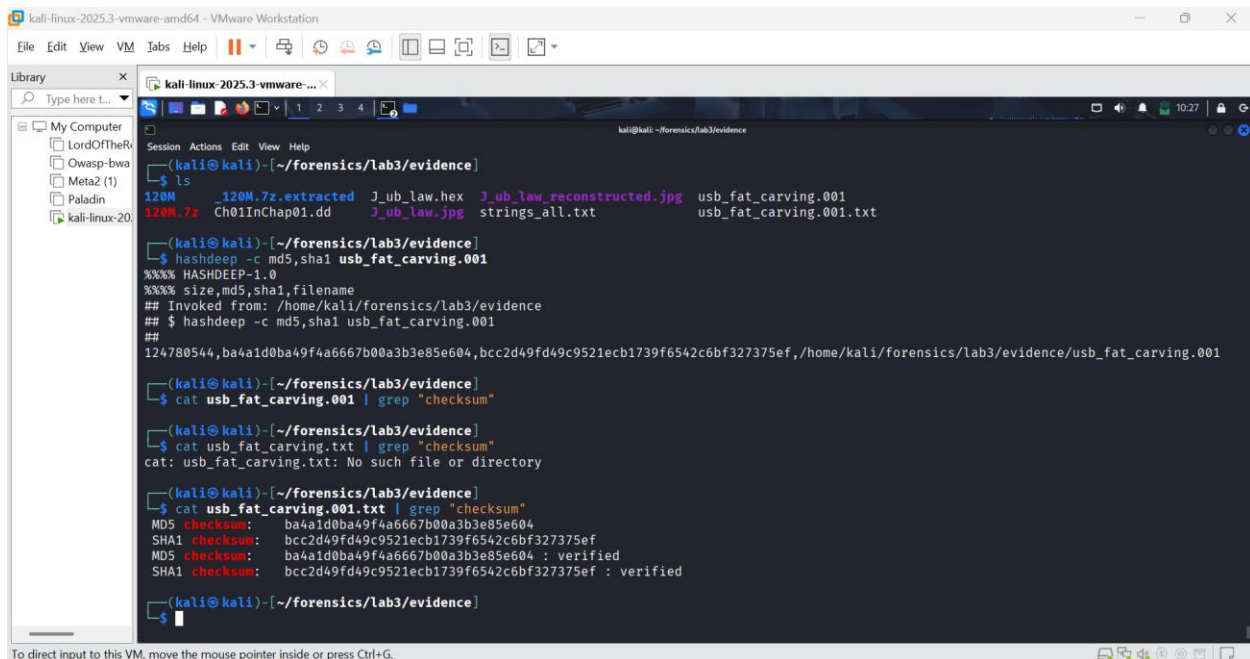


FIGURE 21: I calculated its hashes

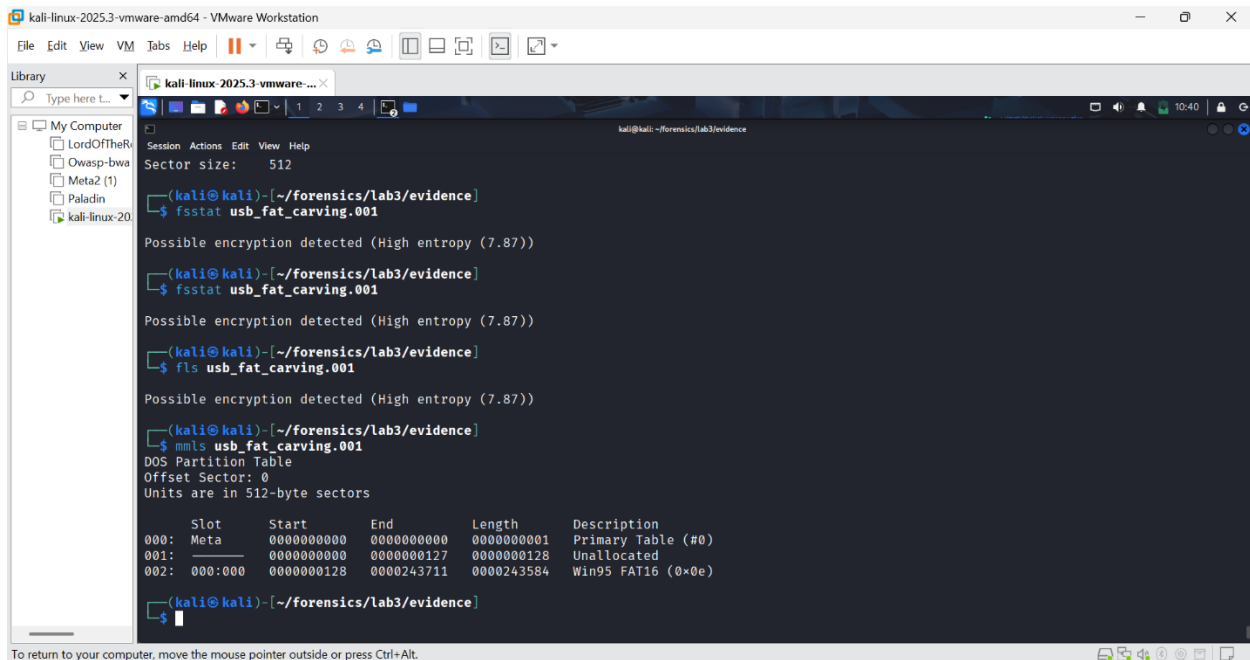


FIGURE 22: I confirmed the appropriate partition offset using mmls which is 128

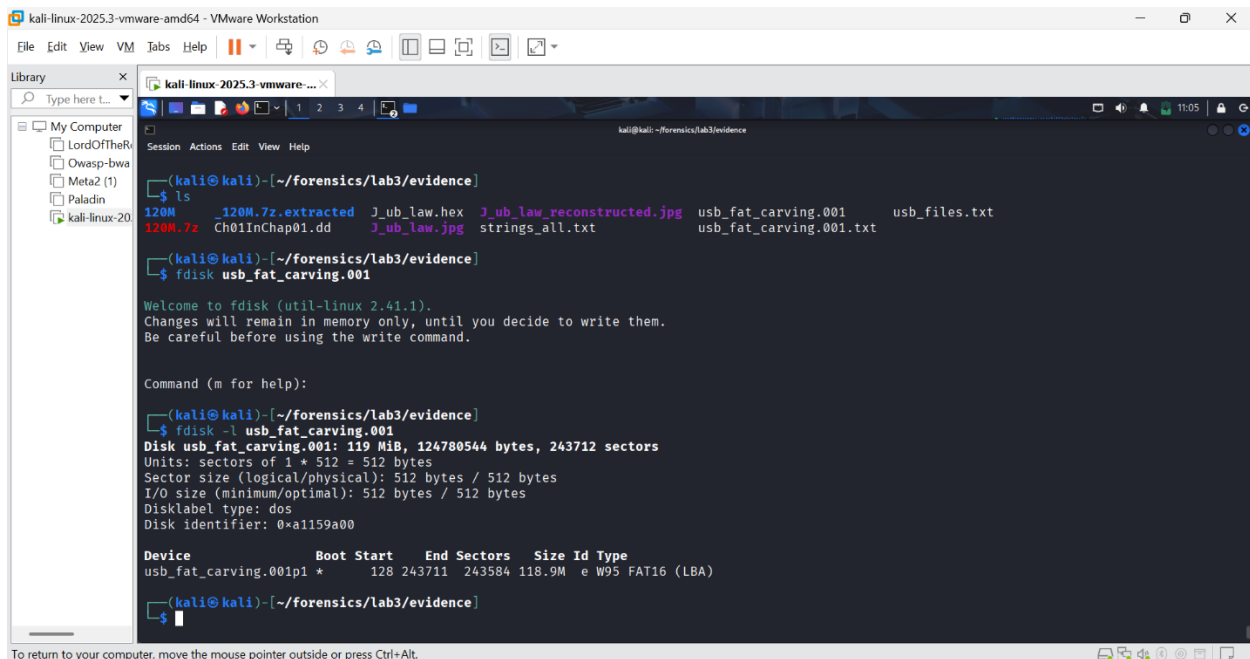


FIGURE 23: I used fdisk to display the partitions

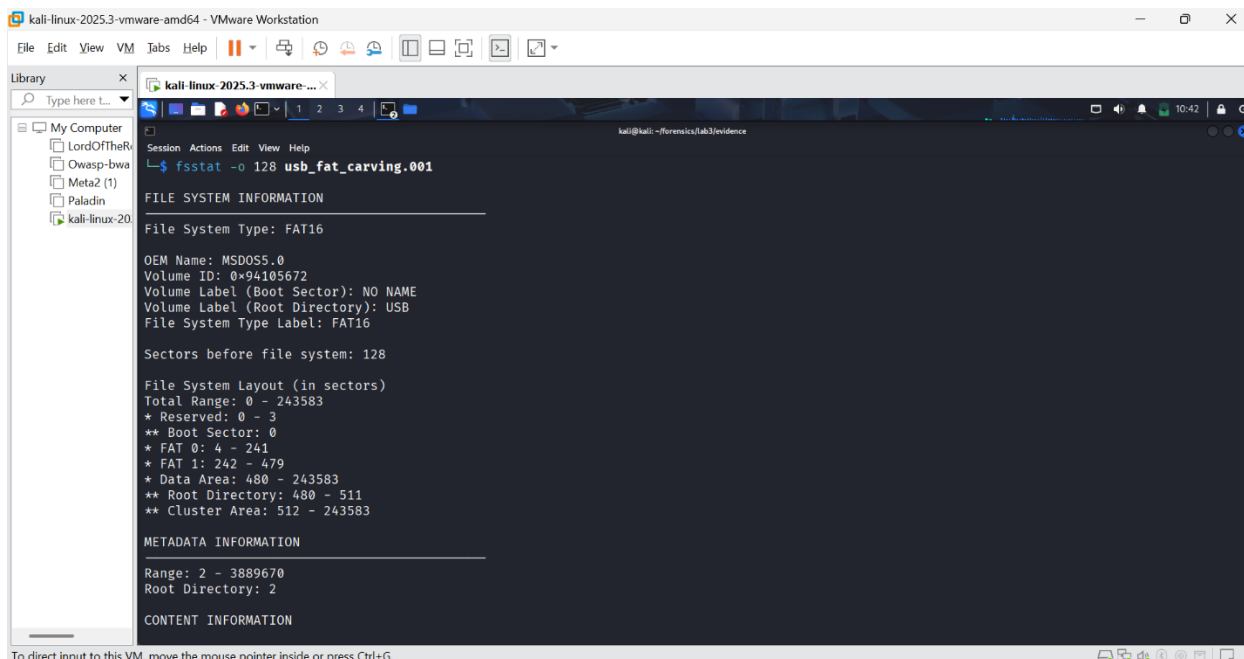


FIGURE 24: File system information

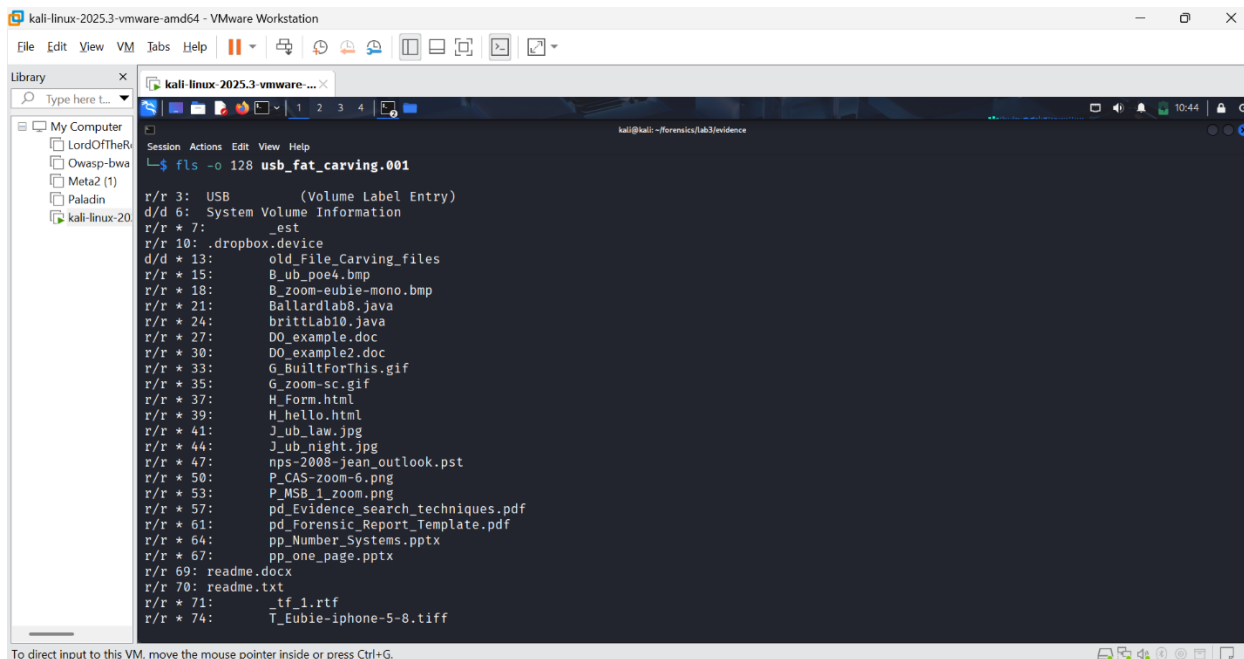
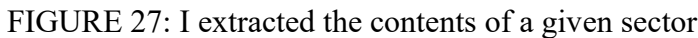


FIGURE 25: File list



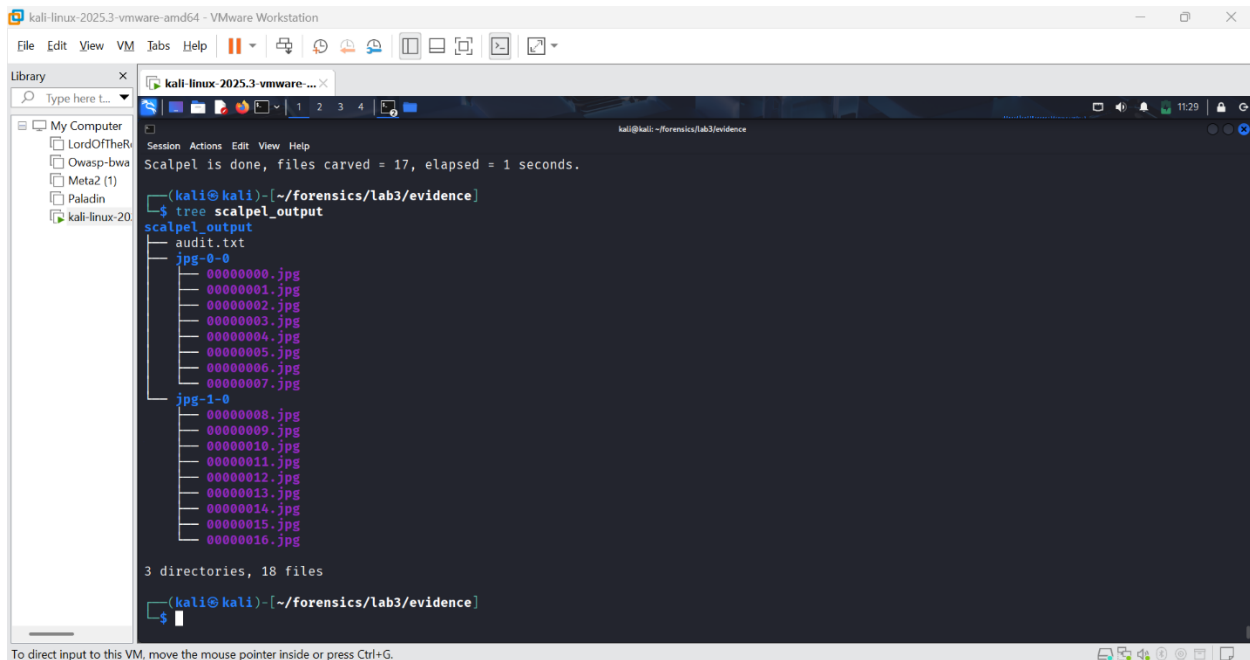


FIGURE 30: Scalpel output showing the carved files

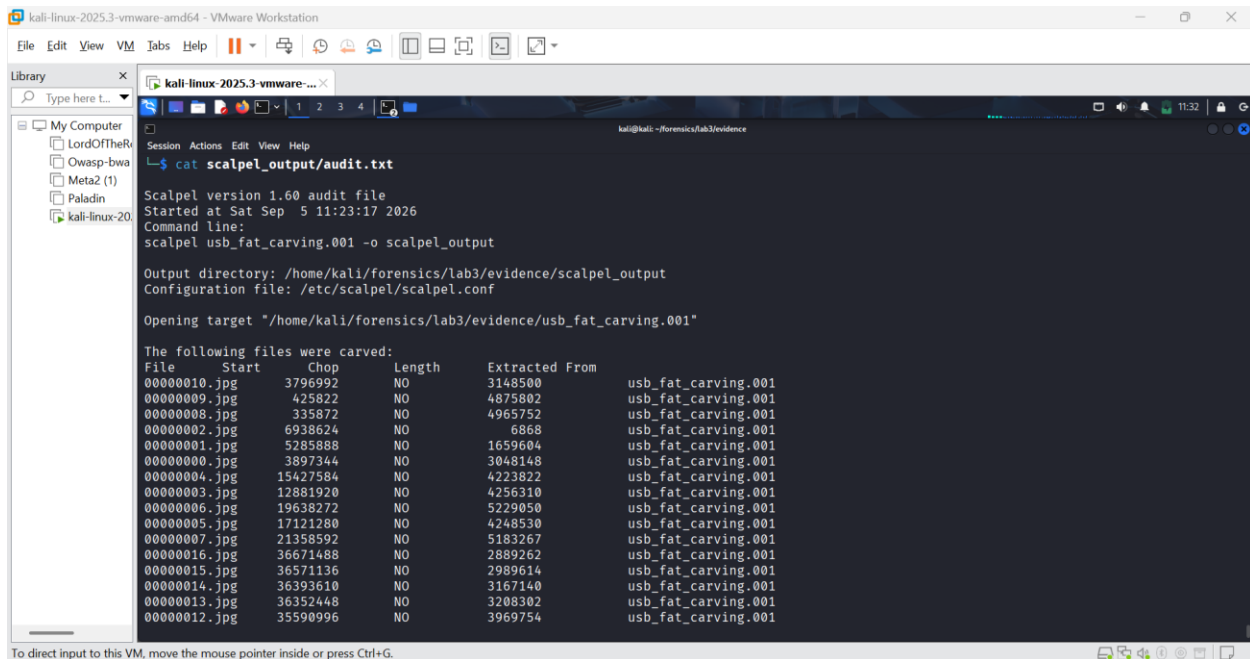


FIGURE 31: Audit log

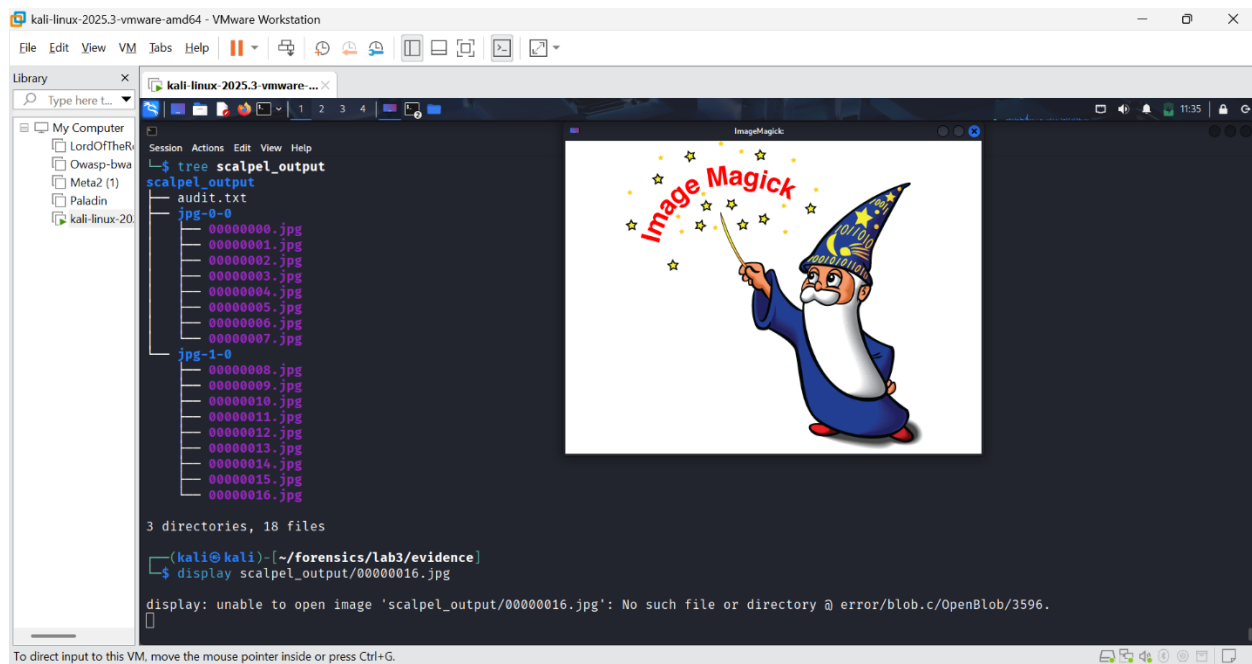


FIGURE 31: I displayed an image